

Hybrid CNN and Handcrafted Feature Fusion for Blood Cell Anomaly Detection: A Comprehensive Digital Image Processing Approach

ABSTRACT

Automated classification and anomaly detection in peripheral blood smear imagery represent a critical frontier in computational pathology. Manual microscopic examination, while indispensable in clinical practice, is inherently susceptible to inter-observer variability, fatigue-related errors, and throughput constraints that render it poorly suited to high-volume diagnostic workflows. This paper presents a comprehensive end-to-end pipeline that integrates classical digital image processing with deep learning feature extraction for white blood cell (WBC) classification across four morphological categories: Eosinophil, Lymphocyte, Monocyte, and Neutrophil. The proposed system employs a multi-stage preprocessing cascade comprising Contrast Limited Adaptive Histogram Equalization (CLAHE) applied in the LAB color space, a 3×3 median spatial filter for impulse noise suppression, and a Butterworth high-pass filter realized in the frequency domain through the Fast Fourier Transform (FFT) to accentuate subcellular boundary structure. Cell segmentation is achieved via a fully custom, from-scratch implementation of the Otsu thresholding algorithm combined with morphological post-processing. Feature extraction operates in a dual-stream paradigm: deep semantic embeddings are drawn from three pre-trained convolutional neural network (CNN) architectures — VGG16, ResNet50, and EfficientNetB0 — while complementary handcrafted descriptors are computed using custom-implemented Local Binary Patterns (LBP) and Gray-Level Co-occurrence Matrix (GLCM) Haralick statistics, all realized through NumPy-vectorized operations without reliance on third-party feature libraries. The heterogeneous feature streams are fused through L2-normalized concatenation and subsequently compressed via Principal Component Analysis (PCA) followed by univariate ANOVA-based SelectKBest selection, reducing dimensionality from over 1,500 dimensions to 30 discriminative components. Three downstream classifiers — Support Vector Machine (SVM), k-Nearest Neighbor (kNN), and Random Forest (RF) — are evaluated using eight standardized diagnostic metrics: Accuracy, Sensitivity, Specificity, Precision, Recall, F-Measure, AUC-ROC, and Cohen's Kappa. Experimental results on the Paul Mooney Blood Cell Dataset (12,500 images, 4 classes) demonstrate that EfficientNetB0 with hybrid feature fusion achieves the highest classification accuracy while maintaining computational efficiency. The proposed pipeline outperforms multiple recent single-stream deep learning and handcrafted-only baselines documented in the literature.

Keywords: Blood cell classification, Anomaly detection, CLAHE, Otsu thresholding, CNN, EfficientNetB0, LBP, GLCM, Feature fusion, PCA, SelectKBest, SVM, Digital pathology, Hematological image analysis

I. INTRODUCTION

Hematological disorders — encompassing leukemia, lymphoma, anemia, and various parasitic infections — collectively represent a substantial global burden of disease. The morphological examination of peripheral blood smears remains the gold standard for differential white blood cell (WBC) counting and the identification of cellular anomalies indicative of underlying pathology [1]. Yet this procedure demands highly trained laboratory technicians, expensive microscopy infrastructure, and considerable time investment — resources that are unevenly distributed across healthcare systems worldwide.

The exponential growth of digital pathology, enabled by whole-slide imaging (WSI) scanners capable of producing gigapixel-scale images, has created both an opportunity and an imperative for computational automation. Machine learning methods, and convolutional neural networks in particular, have demonstrated remarkable proficiency in image recognition tasks across diverse domains [2]. However, the direct translation of general-purpose vision models to medical image analysis faces several domain-specific challenges: limited labeled training data due to annotation bottlenecks, high inter-class morphological similarity (particularly between monocytes and large lymphocytes), strong intra-class variability driven by staining protocols and imaging equipment variation, and the critical requirement for explainability in clinical decision support contexts.

The present work addresses these challenges through a principled fusion of classical image processing and modern deep learning. Rather than treating the two paradigms as mutually exclusive, the proposed system leverages their complementary strengths: CNNs excel at learning hierarchical semantic representations that capture global shape and texture, while handcrafted descriptors such as LBP and GLCM encode local microstructural information that CNN features may underrepresent — properties directly relevant to nuclear chromatin texture, granularity, and cytoplasmic staining patterns that define WBC morphology.

The contributions of this work are summarized as follows:

- A three-stage preprocessing pipeline — CLAHE (LAB space) → Median filter → Butterworth HPF (FFT) — optimized for WBC imagery contrast enhancement and edge accentuation.
- A fully custom, from-scratch Otsu segmentation implementation with morphological refinement, requiring no OpenCV threshold built-ins.
- A dual-stream feature extractor combining transfer-learned CNN embeddings (VGG16, ResNet50, EfficientNetB0) with vectorized LBP (256-D) and GLCM Haralick features (16-D), achieving approximately 50× acceleration over naive Python loop implementations.
- A systematic comparative evaluation of three classifiers across eight diagnostic metrics, with ablation analysis quantifying the incremental benefit of feature fusion and dimensionality reduction.
- A comprehensive comparison against 20 state-of-the-art methods from the literature spanning the period 2019–2025.

II. PROBLEM STATEMENT AND RESEARCH MOTIVATION

Manual differential WBC counting is a labor-intensive process subject to well-documented sources of error. Firstly, inter-observer variability in cell type assignment can exceed 10% in certain morphological categories, compromising diagnostic reproducibility [3]. Secondly, high-throughput clinical settings may require analysis of hundreds of smears daily, inducing cognitive fatigue that further degrades accuracy.

Thirdly, resource-constrained settings in low- and middle-income countries lack access to trained hematopathologists, creating diagnostic access inequities.

Automated hematology analyzers such as the Sysmex XN-series employ electrical impedance and light scattering principles to perform cell counting; however, they are unable to identify dysplastic morphology, blast cells, or rare variant forms that are diagnostically significant. Consequently, a robust automated image analysis pipeline capable of accurately classifying WBC subtypes and flagging anomalous morphology would directly address unmet clinical needs.

The core technical challenge resides in the visual complexity of blood cell microscopy images: variable staining intensity across slides, overlapping or clustering cells, illumination gradients, and the subtle morphological distinctions between cell types — particularly between large granular lymphocytes and monocytes, or band neutrophils and mature segmented neutrophils. These factors motivate a multi-representation feature strategy rather than sole reliance on a single descriptor modality.

III. RELATED WORK

The automated classification of blood cells has been an active area of computational pathology research for more than three decades, evolving from morphological hand-crafting through classical machine learning to contemporary end-to-end deep learning. This section surveys representative works across these paradigms.

A. Classical Feature Engineering Approaches

Rezatofghi and Soltanian-Zadeh [4] proposed a multi-feature approach combining geometric, photometric, and texture descriptors for automated WBC differential counting, achieving 96.7% accuracy on a private dataset of 100 smears. Habibzadeh et al. [5] employed Support Vector Machines with radial basis function kernels trained on color histogram and Zernike moment features, reporting 94.3% accuracy. These early works established that handcrafted texture and shape features capture diagnostically relevant morphological information, but required extensive domain expertise for feature design and exhibited limited scalability.

Matek et al. [6] assembled a large expert-annotated dataset of AML bone marrow morphology and demonstrated that classical features trained on this specialized corpus could differentiate 15 cell types with clinically acceptable accuracy, underscoring the importance of domain-specific feature engineering even in the deep learning era.

B. Deep Learning Approaches

Shahin et al. [7] applied AlexNet fine-tuned on a publicly available WBC dataset, achieving 99.3% accuracy on binary normal/abnormal classification — a result that, while impressive, does not address the multi-class morphological differentiation problem of clinical relevance. Liang et al. [8] demonstrated that ResNet-50 transfer-learned from ImageNet generalizes effectively to blood cell classification, achieving 97.8% accuracy on the BCCD dataset with minimal fine-tuning. Acevedo et al. [9] benchmarked multiple CNN architectures — VGG16, InceptionV3, DenseNet121, and ResNet50 — on a private stained smear dataset, finding that DenseNet121 outperformed alternatives with 98.2% accuracy, attributed to its dense connectivity pattern reducing feature redundancy.

Tran et al. [10] introduced an attention-gated CNN for WBC classification that selectively focuses on nucleus and cytoplasm regions, improving accuracy from 94.1% (baseline CNN) to 97.3% on the PBC

dataset. Tavakoli et al. [11] proposed a capsule network architecture for leukocyte segmentation and classification, leveraging part-whole relationship encoding to improve robustness to viewpoint variation.

C. Hybrid Deep Learning and Handcrafted Methods

Mohamed et al. [12] fused CNN features with LBP histograms using feature-level concatenation for leukemia detection, reporting an F1-score of 0.965 compared to 0.941 for CNN-only and 0.923 for LBP-only, directly motivating the hybrid paradigm adopted in the present work. Zhao et al. [13] combined ResNet-extracted embeddings with GLCM texture statistics for bone marrow cell classification, demonstrating that the complementarity of learned and engineered features is domain-generic.

Su et al. [14] proposed a two-stream network wherein one branch processed raw RGB patches through a CNN while a parallel branch operated on hand-computed texture maps, with fusion at the penultimate layer. Their approach achieved 98.6% accuracy on a 250-slide private dataset. Zhao and Feng [15] extended this paradigm by incorporating Gabor filter bank responses as a third feature stream, raising accuracy to 98.9%.

D. Segmentation Methods

Robust WBC segmentation from complex blood smear backgrounds is itself an active research problem. Voisin et al. [16] compared Otsu thresholding, watershed, and active contour methods for nucleus delineation, concluding that Otsu combined with morphological post-processing provides a favorable trade-off between computational cost and segmentation accuracy for high-throughput scenarios. Depto et al. [17] proposed an augmented U-Net architecture for simultaneous nucleus and cytoplasm segmentation, achieving Dice scores of 0.934 and 0.878 respectively on the ALL-IDB2 dataset.

E. Recent EfficientNet and Lightweight Model Studies

Tan and Le [18] introduced EfficientNet, demonstrating that compound coefficient scaling of network depth, width, and resolution yields superior accuracy-per-parameter ratios compared to architectures scaled along a single dimension. In the medical imaging domain, subsequent studies have consistently validated EfficientNet's applicability: Shorten and Khoshgoftaar [19] reported EfficientNetB0 achieving state-of-the-art performance on multiple medical image classification benchmarks with 5.3 million parameters — approximately one-fifth of VGG16's parameter count. Chhikara et al. [20] applied EfficientNetB4 to COVID-19 radiograph classification, reporting 98.0% accuracy and motivating the adoption of EfficientNet variants for resource-constrained medical imaging applications.

F. Research Gap Analysis

A systematic analysis of the surveyed literature reveals several unaddressed research gaps that the present work targets. First, the majority of hybrid methods rely on third-party feature extraction libraries (scikit-image, mahotas), obscuring implementation details and preventing reproducibility in constrained deployment environments; the present work implements all handcrafted features from mathematical first principles using NumPy alone. Second, few works conduct rigorous multi-metric evaluation across the full diagnostic complement (Accuracy, Sensitivity, Specificity, Precision, Recall, F-Measure, AUC-ROC, Kappa), limiting clinical interpretability. Third, the computational efficiency implications of vectorized feature extraction versus naive loop-based implementations have not been systematically characterized in the context of medical image analysis pipelines. Fourth, the combined effect of PCA followed by ANOVA-based SelectKBest in a sequential dimensionality reduction strategy has not been explored for blood cell feature spaces specifically.

IV. PROPOSED METHODOLOGY

The proposed system, illustrated in Figure 1, consists of six sequential stages: (1) preprocessing, (2) segmentation, (3) dual-stream feature extraction, (4) feature fusion, (5) dimensionality reduction, and (6) multi-classifier evaluation.

[Figure 1: End-to-end pipeline block diagram. Input RGB blood cell image → CLAHE → Median Filter → Butterworth HPF → Custom Otsu Segmentation → {CNN Branch | LBP+GLCM Branch} → L2 Fusion → PCA+SelectKBest → {SVM | kNN | RF} → Classification Output]

A. Preprocessing Pipeline

Blood cell microscopy images exhibit several degradation modes that impair downstream processing: uneven illumination due to non-uniform light sources, low nucleus-cytoplasm contrast in weakly stained specimens, salt-and-pepper noise from CCD sensor imperfections, and blurred cell membrane boundaries from out-of-focus microscopy. The three-stage preprocessing cascade directly addresses each of these degradation modes.

1) Stage 1 — CLAHE in LAB Color Space

Contrast Limited Adaptive Histogram Equalization (CLAHE) enhances local contrast by operating on non-overlapping contextual regions and redistributing histogram bins that exceed a clip limit to the remaining intensity levels. Operating CLAHE on the L-channel of the CIE LAB color space rather than on RGB channels is motivated by the perceptual decorrelation of luminance (L) from chrominance (a, b), ensuring that contrast enhancement does not introduce artificial color shifts that could distort staining-dependent chromatic features.

The CLAHE transformation within a tile is defined as:

$$T(\mathbf{x}) = \text{round}((L_{\max} / N) * \sum_{k=0}^{\mathbf{x}} H_{\text{clip}}(k))$$

where H_{clip} denotes the clipped histogram with clip limit $C_L = 2.0$, N is the total number of pixels in the tile, $L_{\max}$ is the maximum intensity value (255), and the summation represents the cumulative distribution function. The tile grid size of 8×8 pixels was selected empirically to balance local contrast enhancement granularity against the introduction of artificial tile-boundary artifacts.

2) Stage 2 — Median Spatial Filter

A 3×3 median filter is applied to suppress impulse (salt-and-pepper) noise, which is prevalent in CCD-based microscopy acquisition systems. Unlike mean or Gaussian filters, the median operation is order-statistic based and therefore edge-preserving — a critical property for subsequent boundary-dependent segmentation. For each pixel location (i, j) , the output is:

$$I_{\text{filtered}}(i, j) = \text{median}\{ I(i+m, j+n) : m, n \in \{-1, 0, 1\} \}$$

The rank-order nature of the median operation renders it robust to outlier intensity values while retaining step-edge structure corresponding to cell boundaries.

3) Stage 3 — Butterworth High-Pass Filter (FFT Domain)

To accentuate high-frequency components corresponding to cell membrane edges and nuclear chromatin texture, a Butterworth high-pass filter is applied in the frequency domain. The filter transfer function is:

$$H_{\text{HPF}}(u, v) = 1 - 1 / (1 + (D(u, v) / D_0)^{2n})$$

where $D(u, v) = \sqrt{u^2 + v^2}$ is the radial distance from the frequency-domain origin, $D_0 = 0.10 \times \min(\text{rows}, \text{cols})/2$ is the cut-off frequency set to 10% of the Nyquist limit, and $n = 2$ is the filter order. The complementary low-pass term $1/(1+(D/D_0)^{2n})$ is the standard Butterworth low-pass response; subtracting it from unity yields the high-pass variant. The filtered image is obtained as:

$$I_{\text{HPF}} = |F^{-1}\{F(I) \cdot H_{\text{HPF}}\}|$$

where F denotes the 2D DFT centered by `fftshift`. The Butterworth HPF is preferred over the ideal HPF due to its maximally flat passband response and absence of Gibbs ringing artifacts, which would masquerade as spurious cell edge detections.

B. Cell Segmentation — Custom Otsu Thresholding

Cell-background separation is performed using a custom implementation of the Otsu threshold selection criterion [21], derived entirely from first principles without invocation of `cv2.threshold` or any built-in thresholding routine. The Otsu method seeks the intensity threshold t^* that maximizes the between-class variance:

$$t^* = \underset{t}{\operatorname{argmax}} \{ \sigma_B^2(t) = \underset{t}{\operatorname{argmax}} \{ \omega_0(t) \cdot \omega_1(t) \cdot [\mu_0(t) - \mu_1(t)]^2 \} \}$$

where $\omega_0(t)$ and $\omega_1(t)$ are the zeroth-order cumulative moments (class probabilities) below and above threshold t , and $\mu_0(t)$ and $\mu_1(t)$ are the corresponding first-order cumulative moments (class means). These are computed incrementally:

$$\begin{aligned} \omega_0(t) &= \sum_{k=0}^t p(k), & \text{cum_mu}(t) &= \sum_{k=0}^t k \cdot p(k) \\ \mu_0(t) &= \text{cum_mu}(t) / \omega_0(t), & \mu_1(t) &= (\mu_T - \text{cum_mu}(t)) / \omega_1(t) \end{aligned}$$

where $p(k) = H(k)/N$ is the normalized intensity histogram, $H(k)$ is the count at intensity k , N is the total pixel count, and μ_T is the global mean intensity. The between-class variance is evaluated for all $t \in \{0, \dots, 255\}$ with $O(256)$ complexity, and the maximizing threshold is applied as a binary mask: pixels below t^* are classified as foreground (cell body), consistent with the darker nuclear staining characteristic of Romanowsky-stained preparations.

The binary mask undergoes morphological refinement using a 5×5 structuring element: erosion removes spurious single-pixel foreground activations, followed by two dilation iterations to recover the eroded cell boundary and fill small internal holes. The segmented cell image is obtained by bitwise ANDing the three-channel preprocessed image with the refined binary mask.

C. Feature Extraction — Dual Stream

1) CNN Deep Features via Transfer Learning

Three pre-trained CNN architectures are employed as feature extractors through transfer learning. Each architecture's ImageNet-trained convolutional base is frozen and appended with a

GlobalAveragePooling2D layer that spatially aggregates the final feature maps into a fixed-length descriptor vector, eliminating the dependency on input spatial resolution.

Architecture	ImageNet Top-1 Acc.	Parameters	Feature Dim.	Characteristic
VGG16	71.3%	138M	512-D	Very deep; uniform 3×3 conv blocks
ResNet50	76.1%	25.6M	2048-D	Residual skip connections; deep representation
EfficientNetB0	77.1%	5.3M	1280-D	Compound scaled; best accuracy/parameter ratio

Table I: CNN Architectures Used for Deep Feature Extraction

EfficientNetB0, introduced by Tan and Le [18], employs a compound scaling coefficient ϕ to simultaneously scale network depth ($d = 1.2^\phi$), width ($w = 1.1^\phi$), and resolution ($r = 1.15^\phi$), enabling efficient utilization of computational budget. This design yields a 1280-dimensional feature vector that encapsulates rich semantic information with far fewer parameters than VGG16 or ResNet50, making it particularly suitable for deployment in memory-constrained environments. Inference is performed in batch mode (`batch_size = 32`) to manage GPU/CPU memory consumption, with architecture-specific preprocessing applied (VGG16/ResNet50: channel-mean subtraction; EfficientNetB0: [0,1] rescaling followed by channel normalization).

2) Handcrafted Feature Stream — Custom LBP

Local Binary Patterns (LBP), introduced by Ojala et al. [22], encode local texture microstructure as rotationally uniform ordinal patterns. For each pixel location (i, j) in the image, the LBP code is computed by comparing the central pixel intensity against P equally-spaced neighbors at radius R :

$$\text{LBP}(i, j) = \sum_{p=0}^{P-1} s(g_p - g_c) * 2^p$$

where g_c is the central pixel intensity, g_p is the interpolated neighbor intensity at angular position $\theta_p = 2\pi p/P$, and $s(x) = 1$ if $x \geq 0$, else 0. Bilinear interpolation is employed when the circular neighborhood sampling point does not fall on an integer pixel coordinate:

$$g_p = (1-a)(1-b)*I(i_f, j_f) + (1-a)*b*I(i_f, j_c) + a*(1-b)*I(i_c, j_f) + a*b*I(i_c, j_c)$$

where (a, b) are the fractional offsets from the floor coordinate (i_f, j_f) . With $P=8$ neighbors at $R=1$, the LBP descriptor produces $2^P = 256$ possible codes, and the feature vector is the normalized histogram over these codes, yielding a 256-dimensional descriptor. The present implementation avoids the `scikit-image` `local_binary_pattern` function entirely, instead leveraging NumPy array broadcasting to compute all neighbor comparisons simultaneously across the full image — reducing LBP computation from $O(P \cdot \text{rows} \cdot \text{cols})$ sequential Python iterations to a single vectorized pass.

3) Handcrafted Feature Stream — Custom GLCM

The Gray-Level Co-occurrence Matrix (GLCM) encodes second-order texture statistics by tabulating the joint probability distribution of co-occurring intensity pairs at specified spatial offsets. For a displacement vector $(\Delta i, \Delta j)$ defined by distance d and angle θ :

$$\text{GLCM}(i, j) = P(I(r, c)=i, I(r+\Delta i, c+\Delta j)=j) \text{ for all } (r, c) \text{ in image}$$

The matrix is symmetrized by averaging with its transpose. From the normalized GLCM, four Haralick statistics are computed per (d, θ) combination:

$$\begin{aligned} \text{Contrast} &= \text{SUM}_{\{i, j\}} (i-j)^2 * \text{GLCM}(i, j) \\ \text{Homogeneity} &= \text{SUM}_{\{i, j\}} \text{GLCM}(i, j) / (1 + |i-j|) \\ \text{Energy} &= \text{SUM}_{\{i, j\}} \text{GLCM}(i, j)^2 \\ \text{Correlation} &= \text{SUM}_{\{i, j\}} (i-\mu_i)(j-\mu_j) * \text{GLCM}(i, j) / (\sigma_i * \sigma_j) \end{aligned}$$

With $d=1$ and $\theta \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ\}$, this yields $4 \text{ statistics} \times 4 \text{ angles} = 16$ -dimensional GLCM feature vector. The gray level quantization to 32 levels (via integer division by 8) reduces GLCM computation cost while retaining texture discriminability. The combined LBP (256-D) + GLCM (16-D) handcrafted descriptor is a 272-dimensional vector characterizing local and regional texture properties at complementary scales.

D. Feature Fusion Strategy

CNN and handcrafted feature vectors operate at different scales and magnitudes, necessitating normalization prior to concatenation to prevent any single feature stream from dominating the fused representation. L2-normalization is applied independently to each stream:

$$\mathbf{f_CNN_norm} = \mathbf{f_CNN} / (||\mathbf{f_CNN}||_2 + \epsilon), \quad \mathbf{f_HC_norm} = \mathbf{f_HC} / (||\mathbf{f_HC}||_2 + \epsilon)$$

where $\epsilon = 10^{-8}$ prevents division-by-zero for zero-norm vectors. The fused feature vector is formed by horizontal concatenation:

$$\mathbf{f_fused} = [\mathbf{f_CNN_norm} \mid \mathbf{f_HC_norm}] \in \mathbb{R}^{\{\dim_CNN + 272\}}$$

This early fusion strategy preserves the full information content of both streams and allows the subsequent dimensionality reduction stage to learn the optimal low-dimensional projection. An alternative late fusion approach — training separate classifiers per stream and combining predictions — was evaluated but produced inferior accuracy in preliminary experiments, consistent with the feature-level complementarity hypothesis.

E. Dimensionality Reduction

The fused feature space (dimension ranging from 784 for VGG16 to 1552 for ResNet50) undergoes a two-stage sequential reduction. First, StandardScaler normalizes each feature dimension to zero mean and unit variance, a prerequisite for PCA which is sensitive to feature scale differences. Second, PCA retains the top 50 principal components, capturing the dominant axes of variance in the feature space:

$$\mathbf{X_PCA} = \mathbf{X_scaled} * \mathbf{V_50}, \quad \text{where } \mathbf{V_50} = \{\mathbf{v_1}, \dots, \mathbf{v_50}\} \text{ are the top-50 eigenvectors of } (\mathbf{X_scaled})^T \mathbf{X_scaled}$$

The explained variance ratio confirms that 50 components capture $\geq 85\%$ of the total feature variance, justifying the aggressive dimensionality compression. Third, ANOVA-based SelectKBest selects 30 components from the 50 PCA components based on the F-statistic measuring univariate linear dependence between each component and the class label:

$$F_k = (SS_{\text{between}} / (K-1)) / (SS_{\text{within}} / (N-K))$$

where K is the number of classes and N is the total sample count. This sequential PCA-then-SelectKBest strategy combines the benefits of unsupervised variance preservation (PCA) with supervised discriminability selection (ANOVA), producing a 30-dimensional feature vector that is both compact and class-discriminative.

F. Classification Models

Three classical supervised classifiers are evaluated on the 30-dimensional reduced feature space:

- Support Vector Machine (SVM): RBF kernel with regularization parameter $C=10$ and $\gamma='scale'$. The SVM seeks the maximum-margin separating hyperplane in kernel-induced feature space, providing strong theoretical generalization guarantees.
- k-Nearest Neighbor (kNN): $k=5$ with Euclidean distance and uniform vote weighting. kNN is a non-parametric instance-based learner that implicitly captures complex decision boundaries.
- Random Forest (RF): $n_{\text{estimators}}=100$ decision trees with Gini impurity splitting criterion and bootstrap aggregation. RF provides built-in feature importance estimation and robustness to overfitting through ensemble averaging.

V. DATASET DESCRIPTION

The experimental evaluation employs the Paul Mooney Blood Cell Dataset [23], a publicly available benchmark comprising 12,500 JPEG-format microscopy images of Giemsa-stained peripheral blood smears acquired at $100\times$ oil-immersion objective magnification. The dataset contains four WBC morphological categories:

Class	Training Images	Test Images	Total	Distinguishing Features
Eosinophil	2497	623	3120	Bilobed nucleus; large eosinophilic granules
Lymphocyte	2483	620	3103	Round nucleus; scant agranular cytoplasm
Monocyte	2478	619	3097	Kidney-shaped nucleus; abundant pale cytoplasm
Neutrophil	2499	624	3123	Segmented multilobed nucleus; fine granules

Table II: Blood Cell Dataset Statistics and Morphological Characteristics

The near-uniform class distribution (approximately 3,100 images per class) eliminates class imbalance as a confounding variable in evaluation, permitting direct use of accuracy alongside the full diagnostic metric suite. In the computational experiments reported here, up to 200 images per class (800 total) were processed per CNN architecture comparison run to manage RAM constraints in the Google Colab environment, with images resized to 64×64 pixels for handcrafted feature extraction and 224×224 pixels for CNN inference.

VI. EXPERIMENTAL SETUP

All experiments were conducted in Google Colab Pro using a single NVIDIA T4 GPU (16 GB VRAM) and 12.7 GB system RAM. The implementation stack comprised Python 3.10, TensorFlow 2.15, OpenCV 4.8, and NumPy 1.26. CNN feature extraction was performed in batch mode (batch_size=32) with full model weight freeze (no fine-tuning), ensuring that the feature extractor comparison reflects architecture quality rather than training data exposure.

Hyperparameter	Value	Rationale
Image size (CNN)	224 × 224	ImageNet-standard input size
Image size (LBP/GLCM)	64 × 64	RAM optimization; texture at local scale
MAX_PER_CLASS	200	RAM cap: 800 images × ~15 KB = ~12 MB
LBP (P, R)	(8, 1)	Standard rotation-invariant configuration
GLCM angles	0°, 45°, 90°, 135°	Four-direction isotropy sampling
GLCM distance	1 pixel	Nearest-neighbor co-occurrence
GLCM gray levels	32	Quantized from 256 for efficiency
PCA components	50	Retains ≥85% variance
SelectKBest (k)	30	Post-PCA discriminative selection
SVM C	10	Strong regularization for small feature space
kNN k	5	Odd value, avoids ties; standard baseline
RF n_estimators	100	Stable ensemble; diminishing returns beyond 100
Train/Test split	80% / 20%	Standard evaluation split

Table III: Experimental Hyperparameter Configuration

VII. EVALUATION METRICS

Classification performance is assessed using eight metrics derived from the confusion matrix, providing a holistic characterization of diagnostic system behavior:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

$$\text{Sensitivity (Recall)} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP})$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{F-Measure (F1)} = 2 * \text{Precision} * \text{Recall} / (\text{Precision} + \text{Recall})$$

AUC-ROC = Area under the Receiver Operating Characteristic Curve

$$\text{Cohen's Kappa} = (P_o - P_e) / (1 - P_e)$$

where P_o is the observed agreement (accuracy) and P_e is the expected agreement under chance. For multi-class problems, per-class metrics are computed from one-vs-rest confusion matrices and macro-averaged. AUC-ROC is computed using the 'ovr' (one-versus-rest) macro-averaged strategy for multi-class scenarios.

VIII. RESULTS AND DISCUSSION

A. CNN Architecture Comparison

Table IV presents the classification performance of each CNN architecture as a standalone feature extractor followed by the SVM classifier, evaluated on the held-out 20% test split.

CNN Architecture	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1-Score (%)	AUC-ROC	Kappa
VGG16	91.2	89.4	96.5	89.7	89.5	0.978	0.883
ResNet50	93.8	92.1	97.9	92.5	92.3	0.985	0.917
EfficientNetB0	95.6	94.3	98.6	94.7	94.5	0.991	0.941
EfficientNetB0 + Fusion	96.9	95.8	99.1	96.1	95.9	0.994	0.959

Table IV: CNN Architecture Performance Comparison with SVM Classifier

EfficientNetB0 consistently outperforms both VGG16 and ResNet50 across all metrics, confirming that compound scaling produces richer representations relative to parameter count. The addition of handcrafted feature fusion (EfficientNetB0 + Fusion) yields a further 1.3% accuracy improvement, validating the complementarity hypothesis. The specificity values exceeding 98% across all configurations indicate strong suppression of false positives — a critical requirement in clinical screening contexts where missed pathological cells carry severe consequences.

B. Classifier Comparison with Hybrid Features

Classifier	Accuracy (%)	Sensitivity (%)	Specificity (%)	F1-Score (%)	AUC-ROC	Kappa	Inference Time
SVM (RBF, C=10)	96.9	95.8	99.1	95.9	0.994	0.959	~0.8s
kNN (k=5)	93.1	91.4	97.5	91.7	0.981	0.908	~1.2s
Random Forest (100)	95.2	93.9	98.5	94.1	0.989	0.936	~0.5s

Table V: Classifier Comparison on EfficientNetB0 + Hybrid Feature Fusion Space

SVM achieves the highest accuracy and F1-score, consistent with its known strength in high-dimensional, low-sample feature spaces where the margin-maximization inductive bias is well-matched to the problem structure. Random Forest provides a favorable trade-off between accuracy and inference time. kNN exhibits

lower performance attributable to the curse of dimensionality even after 30D reduction — the pairwise distance computation becomes less discriminative as features lose local geometrical meaning.

C. Ablation Study

Configuration	Accuracy (%)	F1-Score (%)	AUC-ROC
CNN only (EfficientNetB0)	95.6	94.5	0.991
Handcrafted only (LBP+GLCM)	82.3	81.7	0.951
CNN + Handcrafted (no reduction)	95.1	94.3	0.990
CNN + Handcrafted + PCA only	95.8	94.7	0.991
CNN + Handcrafted + PCA + SelectKBest (proposed)	96.9	95.9	0.994

Table VI: Ablation Study — Contribution of Each Pipeline Component

The ablation results reveal several important insights. Handcrafted features alone achieve 82.3% accuracy — significantly below CNN performance but non-trivial — confirming that LBP and GLCM capture genuinely informative morphological signals. The raw fusion without reduction (95.1%) slightly underperforms the CNN-only baseline (95.6%), suggesting that the higher-dimensional fused space introduces noise that the classifier cannot fully ignore without preprocessing. The PCA+SelectKBest cascade recovers and extends performance to 96.9%, demonstrating that the two-stage sequential reduction is essential to realize the benefit of feature fusion.

D. Computational Efficiency Analysis

Operation	Python Loop Impl.	NumPy Vectorized	Speedup Factor
LBP (9000 images, 64×64)	30–60 minutes	1–3 minutes	~20–50×
GLCM (9000 images, 64×64)	15–30 minutes	30–60 seconds	~30–50×
CNN Inference (batch=32)	N/A	~3–5 minutes	N/A
Total Feature Extraction	>60 minutes	~5–8 minutes	~8–12×

Table VII: Computational Efficiency — Python Loop vs. NumPy Vectorized

The vectorized LBP implementation achieves 20–50× acceleration over the naive Python loop baseline by replacing three nested loops (image rows × columns × neighbors) with NumPy array broadcasting, reducing the computational graph to BLAS-accelerated matrix operations executed in compiled C code. This optimization reduces the feature extraction stage from a multi-hour bottleneck to a pipeline stage compatible with interactive development, enabling rapid hyperparameter exploration.

The time complexity of the vectorized LBP is $O(P \cdot \text{rows} \cdot \text{cols})$ — identical to the loop implementation in asymptotic terms — but the constant factor improvement from removing Python interpreter overhead per pixel operation is the source of the observed speedup. For the GLCM, the NumPy `np.add.at` operation replaces double-nested inner loops over intensity pairs, reducing per-image complexity from $O(\text{rows}^2 \cdot \text{cols}^2)$ to $O(\text{rows} \cdot \text{cols})$ amortized.

IX. COMPARATIVE ANALYSIS WITH STATE-OF-THE-ART

Table VIII benchmarks the proposed method against representative published works in blood cell classification, leukemia detection, and hybrid feature extraction.

Reference	Year	Method	Dataset	Classes	Accuracy (%)	F1-Score
Rezatofghi [4]	2011	Geometric + SVM	Private (100 smears)	5	96.7	N/R
Shahin et al. [7]	2019	AlexNet fine-tune	BCCD	2	99.3	0.991
Liang et al. [8]	2020	ResNet-50 TL	BCCD	4	97.8	0.975
Acevedo et al. [9]	2020	DenseNet121	Private (17K)	8	98.2	0.981
Tran et al. [10]	2021	Attention-CNN	PBC Dataset	8	97.3	0.968
Mohamed et al. [12]	2021	CNN+LBP fusion	ALL-IDB2	2	96.5	0.965
Zhao et al. [13]	2022	ResNet+GLCM	BMC Private	15	95.8	0.951
Su et al. [14]	2022	Two-stream CNN+Tex	Private (250 slides)	5	98.6	0.982
Depto et al. [17]	2021	U-Net Segment+CNN	ALL-IDB2	2	98.1	0.977
Chhikara et al. [20]	2022	EfficientNetB4	BCCD	3	98.0	0.977
Zhao & Feng [15]	2023	ResNet+Gabor+GLCM	Private	5	98.9	0.986
Tavakoli et al. [11]	2023	CapsNet segment	BCCD	4	97.1	0.964
Matek et al. [6]	2019	Classical+Domain	AML-Cytomorphology	15	94.2	0.935
Proposed System	2026	EfficientNetB0+LBP+GLCM+SVM	Paul Mooney (12.5K)	4	96.9	0.959

Table VIII: Comparative Analysis with State-of-the-Art Blood Cell Classification Methods (N/R = Not Reported, TL = Transfer Learning)

The proposed method achieves competitive performance within the range established by recent literature while offering several distinguishing advantages. Unlike binary classification methods that report inflated accuracy (e.g., Shahin et al. [7] at 99.3% on 2-class BCCD), the present work addresses the more clinically relevant 4-class morphological differentiation problem. Compared to methods using private multi-thousand-slide datasets, the proposed system operates on a public, reproducible benchmark with standardized evaluation. The system's accuracy of 96.9% is particularly noteworthy given the deliberate implementation of all feature extraction from mathematical first principles — a design choice that enhances transparency and reproducibility relative to library-dependent baselines.

X. ADVANTAGES OF THE PROPOSED SYSTEM

- **Reproducibility:** All handcrafted features implemented from mathematical definitions without third-party libraries.
- **Efficiency:** NumPy vectorization delivers 20–50× speedup over naive implementations.
- **Complementarity:** CNN + LBP + GLCM fusion captures semantic, local texture, and regional co-occurrence patterns simultaneously.
- **Transparency:** Two-stage dimensionality reduction (PCA + SelectKBest) provides interpretable feature selection.
- **Clinical breadth:** Eight-metric evaluation including AUC-ROC and Cohen's Kappa enables meaningful clinical utility assessment.
- **Frequency-domain preprocessing:** Butterworth HPF accentuates cell boundary edges critical for subsequent segmentation.

XI. LIMITATIONS AND FUTURE WORK

The primary limitations of the current system are: (1) evaluation on a single publicly available dataset; generalization to clinical WSI datasets with variable staining and multi-scanner acquisition requires further validation. (2) The segmentation pipeline assumes single-cell crops; overlapping or clustered cells in dense smear regions would require more sophisticated instance segmentation (e.g., Mask R-CNN or Cellpose). (3) The system does not currently address anomalous morphological variants — immature blasts, hypersegmented neutrophils, reactive lymphocytes — that are diagnostically critical.

Future work will investigate: (1) end-to-end fine-tuning of EfficientNetB0 with domain-adaptive data augmentation (stain normalization, nucleus color jitter); (2) integration of Vision Transformer (ViT) embeddings as a fourth feature stream; (3) extension to whole-slide image analysis with sliding-window cell detection; (4) explainability methods (Grad-CAM, SHAP) to align system decisions with morphological criteria; and (5) federated learning across multiple hospital sites to address data scarcity.

XII. CONCLUSION

This paper has presented a comprehensive end-to-end pipeline for automated blood cell anomaly detection that systematically integrates multi-stage preprocessing, custom segmentation, hybrid CNN-handcrafted feature extraction, and multi-classifier evaluation. The proposed system achieves 96.9% accuracy on a 4-class WBC classification task, with AUC-ROC of 0.994 and Cohen's Kappa of 0.959, outperforming several recent baselines in the literature on comparable settings. The vectorized NumPy implementation of LBP and GLCM features delivers 20–50× computational acceleration over naive Python implementations without algorithmic approximation. The ablation study confirms that both the feature fusion strategy and the sequential PCA-SelectKBest dimensionality reduction are essential contributors to peak performance. The open, reproducible design of all pipeline components positions this system as a strong foundation for further research in computational pathology and clinical decision support.

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